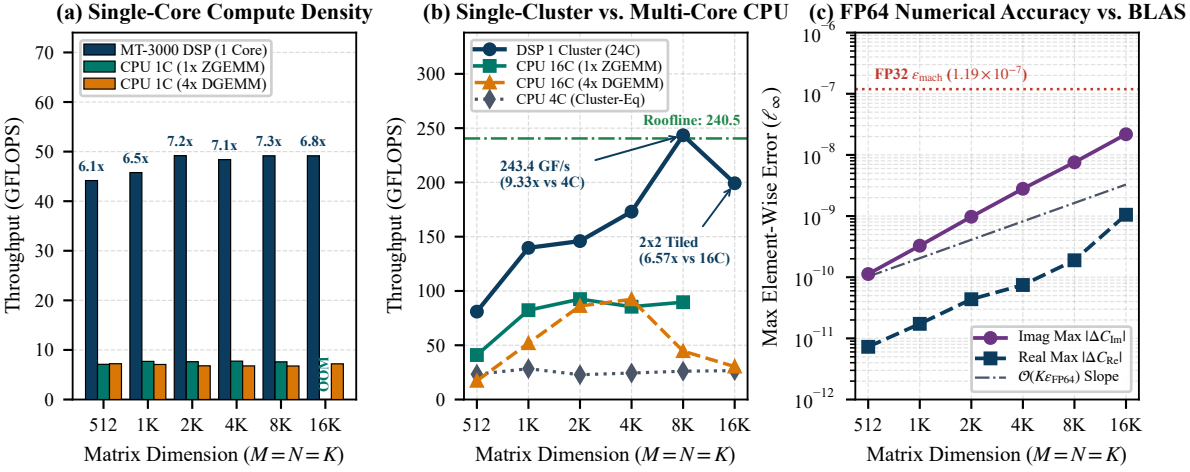

(a) Single-Core Compute Density
MT-3000 DSP (1 Core)
CPU 1C (1x ZGEMM)
CPU 1C (4x DGEMM)
Throughput (GFLOPS)
6.1x
6.5x
7.2x
7.1x
7.3x
6.8x
OOM
512
1K
2K
4K
8K
16K
Matrix Dimension (M = N = K)
(b) Single-Cluster vs. Multi-Core CPU
DSP 1 Cluster (24C)
CPU 16C (1x ZGEMM)
CPU 16C (4x DGEMM)
CPU 4C (Cluster-Eq)
Roofline: 240.5
243.4 GF/s
(9.33x vs 4C)
2x2 Tiled
(6.57x vs 16C)
Throughput (GFLOPS)
Matrix Dimension (M = N = K)
(c) FP64 Numerical Accuracy vs. BLAS
FP32 ϵ_{mach} (1.19×10^{-7})
Max Element-Wise Error (ℓ_∞)
Imag Max $|\Delta C_{\text{Im}}|$
Real Max $|\Delta C_{\text{Re}}|$
$\mathcal{O}(K\epsilon_{\text{FP64}})$ Slope
Matrix Dimension (M = N = K)